

PRODUCT REQUIREMENTS DOCUMENT

Sathvani AVC — Artificial Vocal Cord

Version 1.0 | August 2026 | Confidential

1. DOCUMENT CONTROL

Field	Details
Document Title	Product Requirements Document — Sathvani AVC
Version	1.0
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Author	Sathvani Product & Engineering Team
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Stakeholders	Founder/CEO, Engineering Lead, Clinical Advisor, Regulatory Consultant

2. EXECUTIVE SUMMARY

Sathvani AVC is a mask-style wearable device that restores natural, personalized speech for individuals who have lost their voice due to laryngectomy, vocal cord paralysis, ALS, or related conditions. The device captures physiological signals — vibration (PVDF piezo), acoustic (MEMS microphone), breath pressure, and airflow velocity — and reconstructs phonemes via a cloud-hosted AI pipeline (CNN-BiLSTM-Attention architecture), then synthesizes speech through a personalized TTS engine.

Target Market: 70M+ global voiceless population; initial focus on India's ~10K annual laryngectomy cases plus NGO/rehabilitation partnerships. TAM \$4.3B (global AAC & SGD), SAM \$460M (APAC), SOM \$1.5M (Year 1–3 India).

3. PROBLEM STATEMENT

Over 70 million people worldwide cannot speak. Current solutions force unacceptable tradeoffs:

Solution	Critical Failure
Electrolarynx	Robotic monotone; held against throat; socially stigmatizing; no personalization
AAC Tablets/Apps	Requires fine motor skills; too slow for real conversation; stigmatizing in public
Speech Therapy	Limited outcomes for total vocal cord loss; months of effort for marginal gains
TEP (Tracheoesophageal Puncture)	Invasive surgery; infection risk; only viable post-laryngectomy; not personalized

Core Gap: No existing non-invasive device converts physiological speech-production intent (breath, vibration, pressure, airflow) directly into natural, personalized, real-time speech without surgery or a robotic voice box.

4. PRODUCT VISION & OBJECTIVES

Vision: Restore the dignity of human voice for every person who has lost it.

Product Objectives (12-month horizon):

- Build and validate a functional wearable prototype with 4-sensor acquisition
- Achieve >85% phoneme reconstruction accuracy on real patient data (laryngectomy / paralysis)
- Deliver sub-500ms end-to-end latency (sensor input → synthesized speech output)
- Complete 10-patient clinical pilot at AIIMS Bibinagar
- File complete patent specification and initiate CDSCO Class A/B regulatory pathway
- Secure FDA pre-submission meeting and CDSCO manufacturing license

5. TARGET USERS & PERSONAS

Primary User — The Voiceless Patient

Post-laryngectomy patients (age 45–75), vocal cord paralysis patients, and progressive neurological condition sufferers (ALS, Parkinson's). Needs: effortless, hands-free, natural-sounding speech in social and clinical settings. Cannot type or hold devices for extended periods.

Secondary User — Caregiver & Clinician

ENT surgeons, speech pathologists, and rehabilitation centre staff. Needs: objective usage data, calibration tools, training mode, and integration with existing therapy workflows.

Tertiary User — Payer & Regulator

Government ADIP scheme administrators, insurance providers, hospital procurement. Needs: demonstrated clinical efficacy, regulatory compliance (CDSCO/FDA), and cost-effectiveness vs. existing alternatives.

6. FUNCTIONAL REQUIREMENTS

6.1 Hardware — Wearable Device

ID	Requirement	Priority	Acceptance Criteria
FR-HW-001	PVDF piezoelectric sensor for vibration capture (primary signal, ~6500Hz, 20kPa)	P0	Signal confirmed on ESP32-S3; 16-bit 16kHz sampling
FR-HW-002	MEMS microphone (I2S) for acoustic capture	P0	Driver confirmed on ESP32-S3; 16-bit 16kHz sampling
FR-HW-003	Pressure sensor (I2C) for breath pressure	P0	Range 0-10 kPa; pending bench test on patient-mimic
FR-HW-004	Airflow velocity sensor (I2C) for articulatory airflow	P0	Integrated in prototype; range 0-30 m/s
FR-HW-005	32-bit ADC for piezo signal acquisition	P0	≥24-bit effective resolution; sampling ≥1kHz
FR-HW-006	ESP32-S3 preprocessing unit (sync + buffer + WiFi relay)	P0	Real-time multi-sensor sync; UDP streaming to cloud
FR-HW-007	Rechargeable battery (≥6hr continuous use)	P1	Validated via discharge cycle testing
FR-HW-008	Mask-style ergonomic form factor (<150g)	P1	User comfort score ≥4/5 in pilot
FR-HW-009	LED status indicator + haptic feedback	P2	Visual/audio feedback for connection/ battery/ error states
FR-HW-010	BLE pairing for companion mobile app	P2	Stable BLE 5.0 connection; <2s pairing time

6.2 AI/Cloud Pipeline — Signal Processing & Phoneme Reconstruction

ID	Requirement	Priority	Acceptance Criteria
FR-AI-001	Extract 13 engineered acoustic descriptors per phoneme (Distance RMS, SPL, Pressure, Velocity, Duration, Energy Ratio, Duration, etc.)	P0	Descriptors confirmed on test set; 100% accuracy
FR-AI-002	Gaussian noise augmentation (40x per sample, $\sigma=0.04$ mean absolute magnitude)	P0	Implemented augmentation pipeline; prevents overfitting on sn
FR-AI-003	CNN feature extractor: 1D conv stack 64→128→256 channels	P0	BatchNorm + GELU + Dropout per block
FR-AI-004	BiLSTM temporal modeling: 3 layers, 256 hidden units, 50 directions	P0	Captures forward/backward phonetic dependencies
FR-AI-005	Multi-head self-attention: 4 heads with residual connections + LayerNorm	P0	Contextual refinement of latent representations
FR-AI-006	Classification head: softmax over phoneme classes with label smoothing	P0	Weighted cross-entropy; inverse-frequency class weights
FR-AI-007	AdamW optimizer + cosine annealing warm restarts	P0	LR=2e-3, weight decay=1e-4, batch=32
FR-AI-008	CMU phonetic dictionary mapping for word candidate generation	P0	Dynamic segmentation + edit-distance recovery
FR-AI-009	Language model re-ranking (GPT-2 or equivalent) for sentence plausibility	P0	Scoring -Loss_LM(sentence); top-1 selection
FR-AI-010	TTS synthesis: FastSpeech2 + HiFi-GAN for natural voice output	P0	MOS ≥3.5; latency <200ms per utterance
FR-AI-011	Personalized voice cloning from pre-loss recordings	P1	Speaker embedding adaptation; 5-min sample sufficient
FR-AI-012	KNN centroid-based unseen phoneme generalization	P1	Fallback inference for out-of-vocabulary phonetic patterns

6.3 Software — Companion App & Cloud Platform

ID	Requirement	Priority	Acceptance Criteria
FR-SW-001	Real-time audio playback of synthesized speech on mobile device	P0	Latency <500ms end-to-end

ID	Requirement	Priority	Acceptance Criteria
FR-SW-002	User voice profile creation and management	P0	Upload pre-loss audio; generate speaker embedding
FR-SW-003	Device pairing, calibration, and firmware OTA updates	P0	BLE + WiFi dual-mode; secure OTA
FR-SW-004	Usage analytics dashboard for clinicians	P1	Daily usage time, phoneme accuracy trends, error logs
FR-SW-005	Offline mode: cached TTS for common phrases	P1	50 pre-loaded phrases; no cloud needed
FR-SW-006	Multi-language support (Hindi, Telugu, English initial)	P1	Phoneme inventory covers target language phones
FR-SW-007	Subscription management (₹500–1,500/month SaaS tier)	P2	In-app purchase + hospital billing integration
FR-SW-008	Anonymized data export for research partners	P2	HIPAA/GDPR-compliant de-identification

7. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Target
Performance	End-to-end latency (sensor → speech)	< 500ms (P0); < 300ms (P1 stretch)
Performance	Phoneme reconstruction accuracy	> 85% on real patient data (P0); > 90% (P1)
Performance	TTS naturalness (MOS)	≥ 3.5 / 5.0
Reliability	Device uptime	≥ 99.5% during 6hr session
Reliability	Cloud API availability	≥ 99.9% uptime SLA
Security	Data encryption at rest	AES-256
Security	Data encryption in transit	TLS 1.3
Security	Patient data compliance	HIPAA (US), DPDP Act 2023 (India), GDPR (EU)
Scalability	Concurrent users (Year 1)	Support 2,300+ active devices
Scalability	Cloud inference throughput	≥ 100 req/sec per inference node
Usability	First-time setup time	< 15 minutes
Usability	Device weight	< 150 grams
Regulatory	India CDSCO	Class A/B medical device license
Regulatory	US FDA	510(k) or De Novo pathway
Regulatory	ISO 13485	Quality management system certification

8. USER STORIES

As a...	I want to...	So that...	Priority
Post-laryngectomy patient	speaking naturally without holding a device	I can have normal conversations in public	P0
Post-laryngectomy patient	hear my own pre-surgery voice	I retain my identity and emotional connection	P1
ALS patient with hand tremors	communicate without typing or pressing buttons	I can communicate despite motor limitations	P0
Speech pathologist	view patient usage and accuracy data	I can tailor therapy and track progress	P1
Hospital procurement officer	purchase devices in bulk at ≤ ₹65K/unit	we can offer it under ADIP scheme	P1
ENT surgeon	see clinical validation data	I can confidently recommend it to patients	P0

9. SUCCESS METRICS & KPIs

Technical KPIs (6-month)

- Phoneme accuracy ≥ 85% on 10-patient pilot dataset
- End-to-end latency < 500ms (p95)
- Device battery life ≥ 6 hours continuous

- Cloud inference cost < █3 per 1000 utterances

Business KPIs (12-month)

- 2,300+ devices sold/distributed (Year 1 SOM target)
- 500+ active SaaS subscribers (█500–1,500/month)
- AIIMS pilot completion with IRB approval
- CDSCO manufacturing license obtained
- Complete patent specification filed

10. RISKS & MITIGATIONS

Risk	Impact	Likelihood	Mitigation
Research paper test/dist files were identical — 100% accuracy	High	High	Hybrid generalization: re-align target data; patient data; redesign train/test split with
Sensor signal quality insufficient in acoustic phonetic patients	Critical	Medium	Early bench testing with patient-mimic phantoms; partner with AIIMS for Day
Regulatory delays (CDSCO/FDA)	High	Medium	Engage regulatory consultant now; file provisional → complete patent; pursu
Competitor enters with larger funding (Medtronic, Stryker, etc.)	Medium	Medium	Secure IP via complete patent; build clinical data moat through AIIMS partne
Cloud latency unacceptable on 4G/weak networks	Critical	Medium	Edge quantization of CNN-BiLSTM (TFLite/ONNX); deploy edge TTS cache;
TTS personalization requires high-quality voice recordings	Medium	Low	Offer generic high-quality voice models as default; partner with voice banking

11. OPEN QUESTIONS

- What is the minimum viable sensor subset if pressure/airflow prove unreliable in patients? (Fallback: piezo + mic only?)
- Can the CNN-BiLSTM-Attention model be quantized to run partially on ESP32-S3 for edge inference?
- What is the regulatory classification in EU (MDR Class I/IIa)?
- Should GPT-2 be replaced with a smaller, open-source LM (e.g., LLaMA-3-1B) for cost and licensing reasons?
- What is the go-to-market priority: B2B hospital bulk (ADIP) vs. B2C D2C portal?

12. APPENDICES

Appendix A: Research Paper Summary — 'Deep Acoustic-Phonetic Reconstruction Framework Using CNN-BiLSTM Attention Networks with Explainable AI' (Doc 1). Key findings: RMS amplitude dominates SHAP importance (0.4134); 13-descriptor feature space enables near-perfect accuracy on clean data; CMU dict + GPT-2 pipeline validates sentence reconstruction.

Appendix B: Pitch Deck Summary — Sathvani AVC (Doc 2). Seed ask: █30–50L; target price █65K; roadmap: 2025 hardware → Q1-Q2 2026 prototype → Q3-Q4 2026 clinical validation → 2027–2028 FDA/CDSCO approval.

Appendix C: Competitive Landscape — Electrolarynx (robotic), Voiceitt (dysarthric speech clarity, not aphonic), Tobii Dynavox (eye-gaze AAC), TEP (surgical). AVC occupies unique white space: non-invasive physiological-signal-to-speech for zero-voice patients.

— *End of PRD* —